



SIMATS
ENGINEERING



SIMATS
Saveetha Institute of Medical And Technical Sciences
(Declared as Deemed to be University under Section 3 of UGC Act 1956)

CSA-0508 Database Management Systems

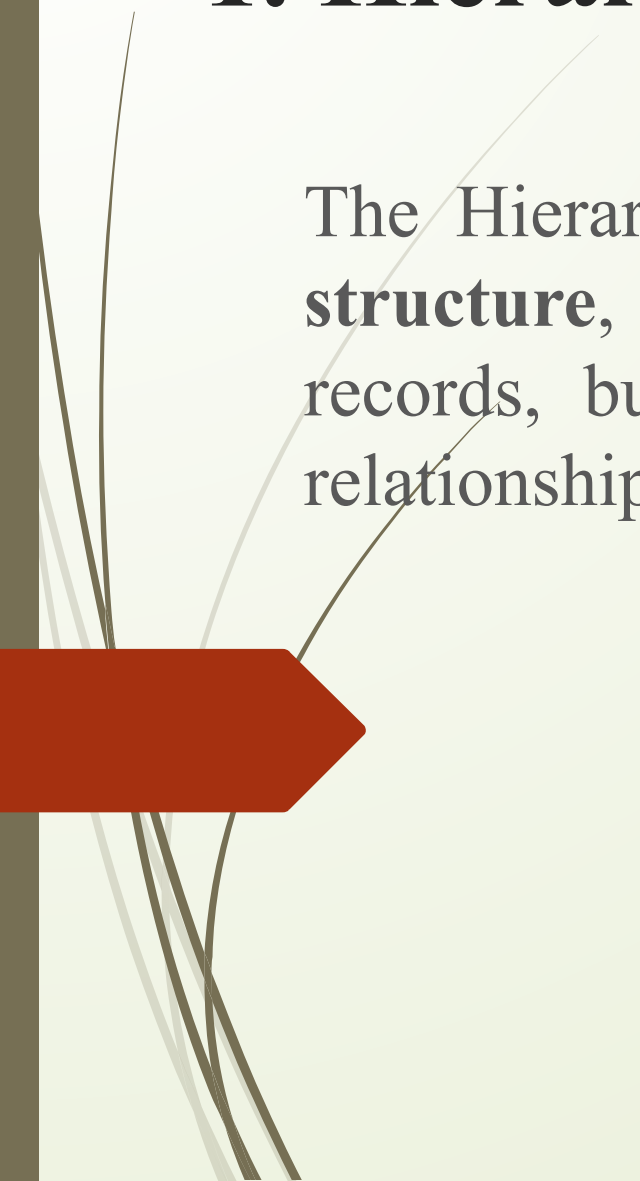
Different Data Models (Hierarchical, Network, Relational)

Guided By:
Dr. Ayyappan G

Done By:
P. Hemachandran(192525440)

1. Hierarchical Data Model

The Hierarchical Data Model organizes data in a **tree-like structure**, where each parent record can have many child records, but each child has only one parent (one-to-many relationship).





Characteristics

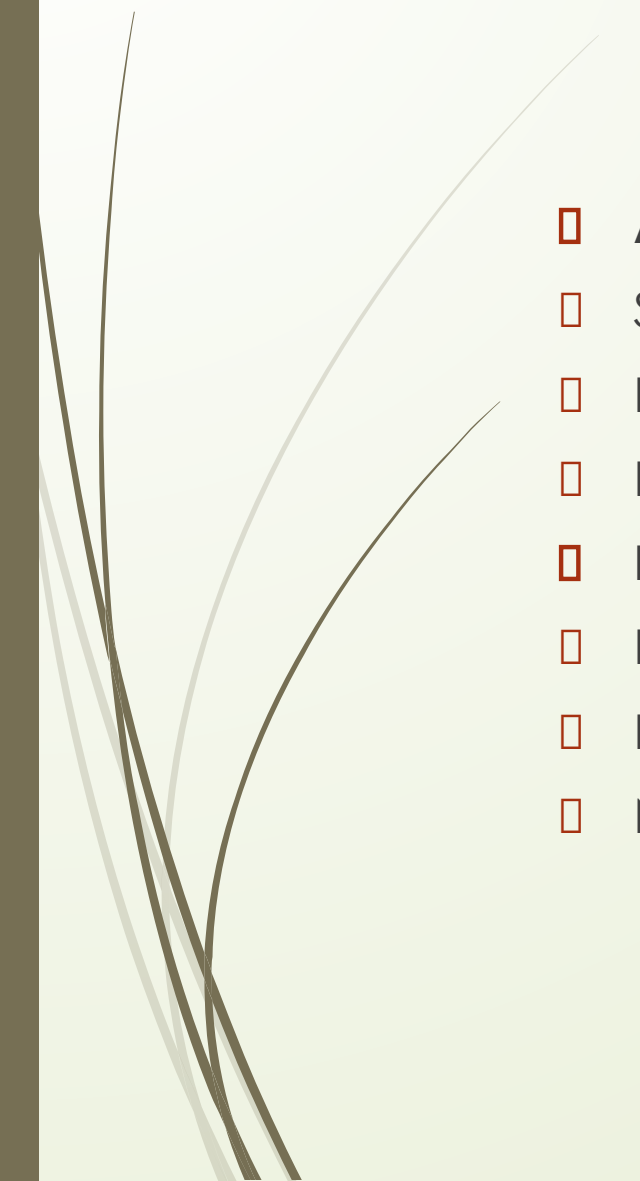
- Tree structure.
- One parent can have many children.
- Each child has only one parent.
- Fast data retrieval.
- Difficult to represent many-to-many relationships.

□ **Industry Use Cases**

- Organization hierarchy
- File systems (Windows/Linux directories)
- XML documents
- Company organizational charts



Advantages/Disadvantages



□ **Advantages**

- Simple and easy to understand.
- High performance for hierarchical data.
- Fast searching

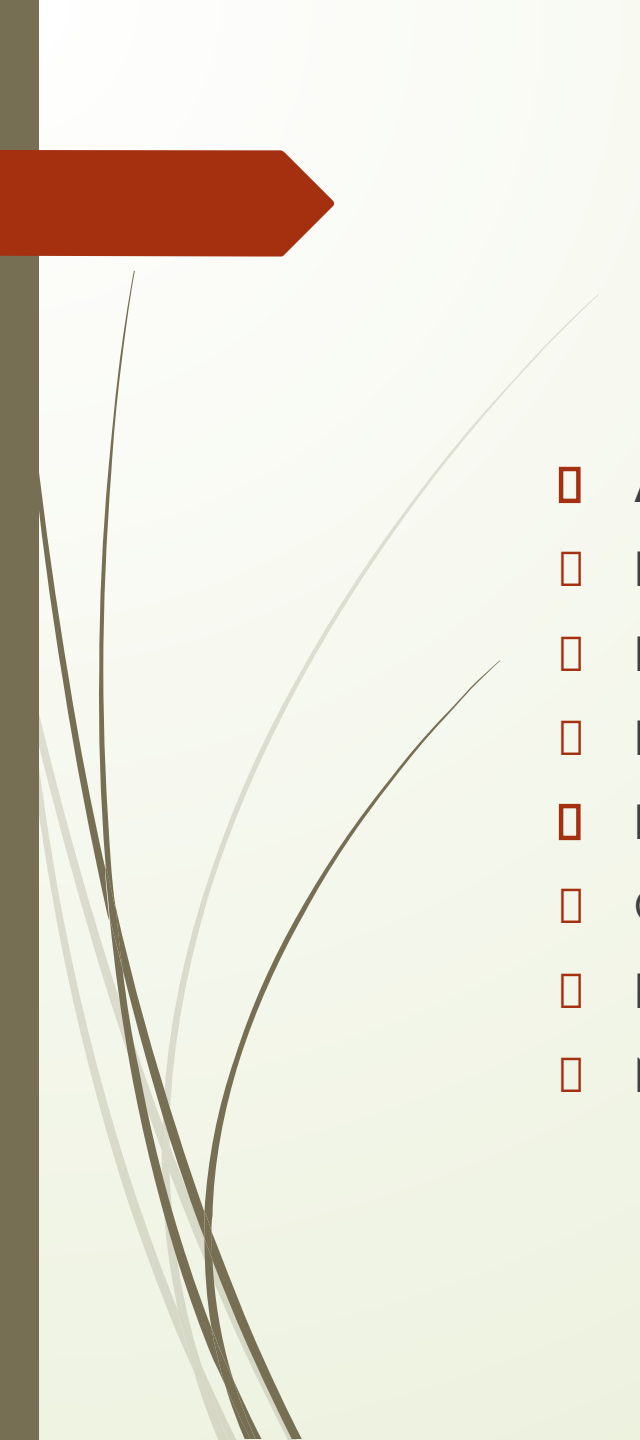
□ **Disadvantages**

- Data redundancy.
- Difficult to modify.
- No many-to-many relationship.



2. Network Data Model

- ❑ The Network Data Model allows **many-to-many relationships**, where a child can have multiple parents.
- ❑ **Characteristics**
 - ❑ Graph structure.
 - ❑ Multiple parent and child relationships.
 - ❑ Supports many-to-many relationships.
 - ❑ Uses pointers (links).



❑ **Advantages**

- ❑ Less redundancy.
- ❑ Faster access than hierarchical.
- ❑ Flexible relationships.

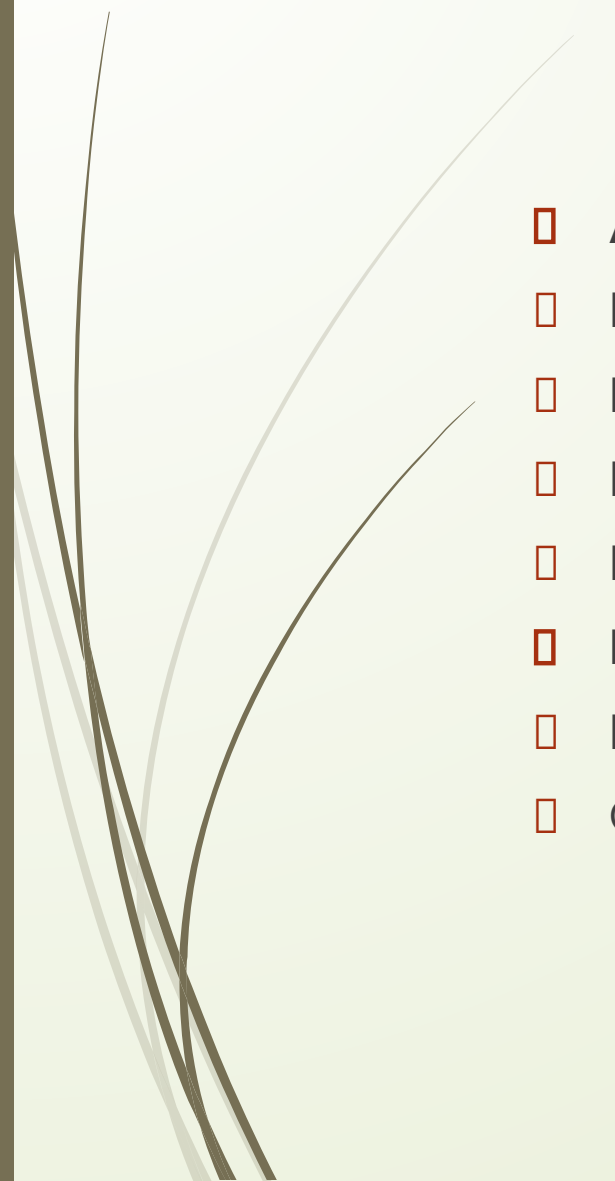
❑ **Disadvantages**

- ❑ Complex design.
- ❑ Difficult to maintain.
- ❑ Navigation is complicated.



3. Relational Data Model

- ❑ The Relational Data Model stores data in the form of **tables (relations)** consisting of rows and columns
- ❑ **Characteristics**
 - ❑ Data stored in tables.
 - ❑ Uses Primary Key and Foreign Key.
 - ❑ Supports SQL.
 - ❑ Reduces redundancy through normalization.



❑ **Advantages**

- ❑ Easy to understand.
- ❑ High data integrity.
- ❑ Flexible querying using SQL.
- ❑ Easy maintenance.

❑ **Disadvantages**

- ❑ Performance may decrease with many joins.
- ❑ Complex databases require optimization.